9/1/2026

N + - Network Fundamentals & Building Networks.

Network Topologies.

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TASK 1.

❖ **Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.**

➢

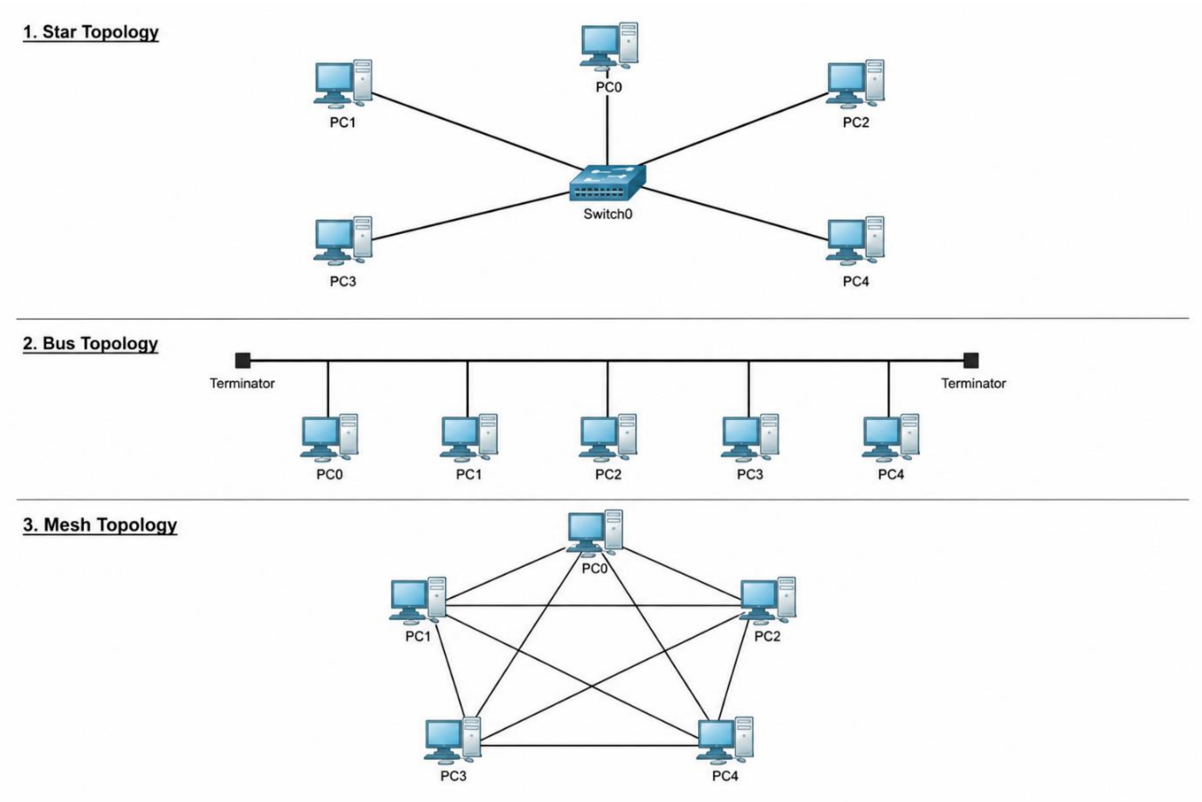

Figure : Star, Bus and Mesh Network Topologies.

Star Topology

- ✓ **Real-world example:** WhatsApp
  In a star topology, all devices are connected to a central device such as a switch. It is useful for managing communication between multiple devices from one central point.

➢ Bus Topology

- ✓ **Real-world example:** Flipkart
  In a bus topology, all devices share a single main communication line. It is simple and requires less cabling, making it suitable for small networks.

➢ Mesh Topology

- ✓ **Real-world example:** IPL Streaming
  In a mesh topology, devices have multiple connections with each other. This provides alternate paths for data, making the network more reliable if one connection fails.

TASK 2.

❖ **Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).**

➢ **Comparison of Network Topologies:**

Topology	**Pros (4)**	**Cons (4)**	**Best Use Case**	**Example**
Star	• Easy to set up • Easy to manage • One device failure does not affect others • Easy to expand	• Depends on central device • Requires more cables • Higher cost • Central device failure affects all	Offices, schools, labs	Instagram
Bus	• Simple design • Uses less cable • Low cost • Easy to install	• Backbone failure affects all • Slows with more devices • Difficult to troubleshoot • Limited network length	Small networks	Zomato
Ring	• Organised data flow • Equal access to devices • Fewer data collisions • Good for steady traffic	• One break can affect the network • Difficult to expand • Harder to troubleshoot • High traffic can reduce speed	Networks with fixed data flow	Paytm

Mesh	• Highly reliable • Multiple paths for data • One link failure does not stop network • Good for critical communication	• Expensive to install • Requires more cables/hardware • Complex setup • Difficult to maintain	Critical and large networks	WhatsApp

TASK 3.

❖ **Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.**

➢ **Network topology for a multiplayer mobile game:**

- I would choose **Mesh Topology** for a multiplayer mobile game because it provides multiple paths for data communication. This helps reduce delays and keeps the game running even if one connection fails. It offers high speed and reliability, but the setup and maintenance cost is higher. For a multiplayer game, this extra cost is justified for a smooth and stable gaming experience.

TASK 4.

❖ **Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines, Consider factors like cost, ease of troubleshooting, and future expansion.**

➢ **Recommended Network Topology:**

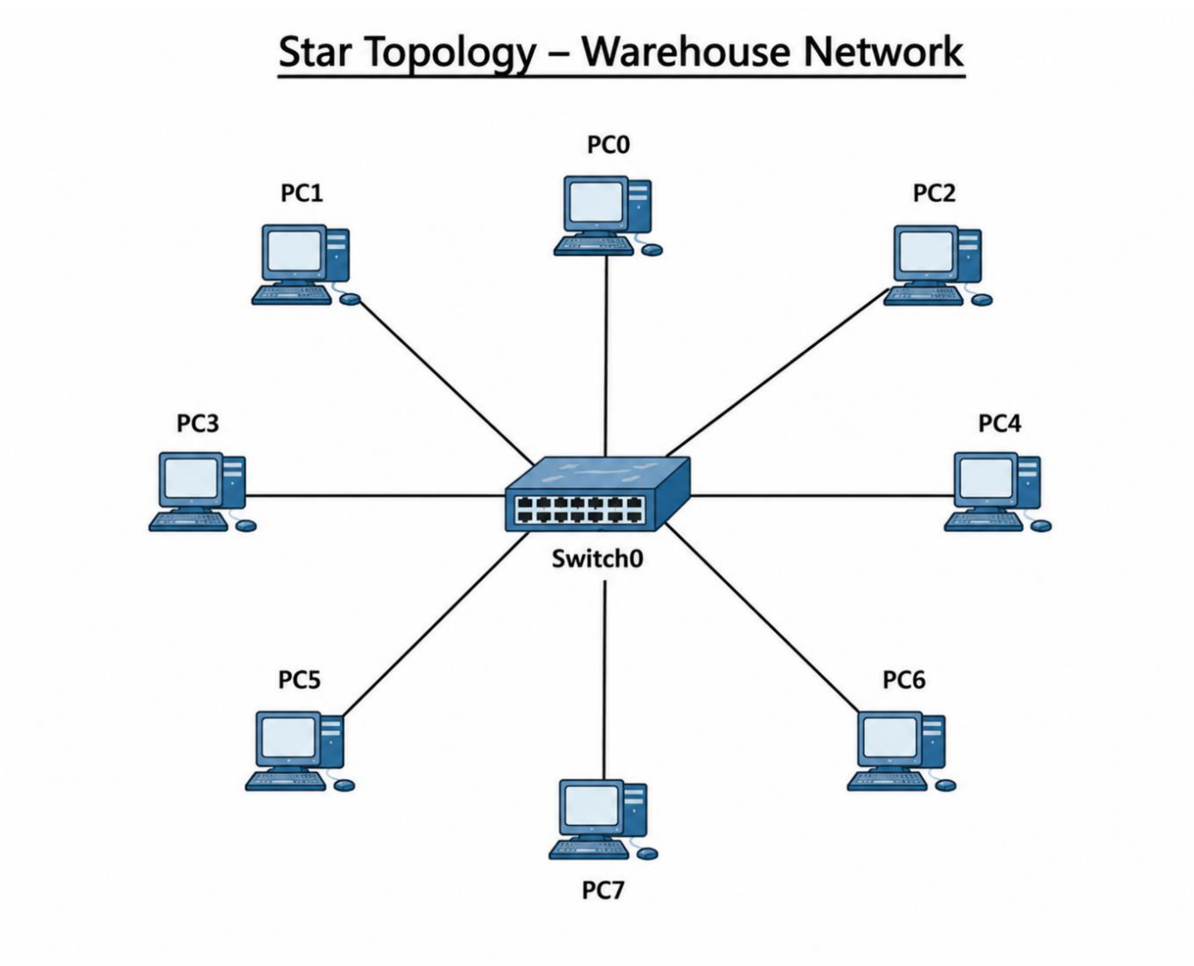

Figure : Star Topology for a Small Warehouse Network.

Explanation :

- I would recommend **Star Topology** for the warehouse because it is cost-effective and easy to manage for 8 computers. Troubleshooting is simple since each computer has a separate connection to the central switch. It also allows easy future expansion by adding more computers to the switch.